

An introduction to SOLPS

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- in which we introduce you to the SOLPS suite of codes

Overview of this course

- Introduction
 - Coster
- DG
 - Coster
- Carre
 - Bonnin
- aggj
 - Coster
- B2.5
 - Coster
- Eirene
 - Reiter
- B2.5-Eirene
 - Bonnin
- DN & limiter cases
 - Bonnin
- Analysis tools
 - Coster, Wischmeier
- SOLPS6.0
 - Bonnin
- Edge2d tools relevant to SOLPS
 - Corrigan
- User impressions
 - Wischmeier
 - Reiser
 - Tsitrone

The SOLPS suite of codes consist of

- a number of different codes

B2 a multi-fluid plasma code, originally written by Bas Braams, but substantially modified since

Eirene a Monte-Carlo kinetic neutrals code, written by Detlev Reiter

B2-Eirene coupling of B2 and Eirene

Sonnet a gridding code written by Hans-Peter Zehrefeld

Carre a gridding code written by Richard Marchand

DG a graphical tool used for developing and modifying plasma devices and plasma grids, as well as producing input files for some of the other codes, written by Andrei and Konstantin Kukushkin

b2plot used for plotting results from simulation runs, written by David Coster and subsequently modified by others

various scripts

- that are used to simulate transport in the edge plasma of tokamaks (and toroidally averaged stellarators)

Currently 3 versions of SOLPS exist

SOLPS4.0 in widespread use

- uses the older B2 code
 - potential equation eliminated from the set of equations
 - no drifts or currents
 - only a simple neutral model contained in B2
- well validated
- currently used for ITER simulations
- maintenance only, no (or only limited) new development
- not recommended that new people use
- only old work should be completed with this version

SOLPS5.0 current version

- uses the newer B2.5 code
 - solves the potential equation
 - has drifts and currents
 - * though drift terms are usually switched off
 - comprehensive fluid neutral model
- use is increasing
- under continuing development
 - particularly in the area of drifts and currents

SOLPS6.0 in development

- grid refinement
- the subject of a complete presentation by Xavier Bonnin

How big is the code?

B2 162 subroutines, 87944 lines of FORTRAN 77 code

B2.5 439 subroutines, 87069 lines of FORTRAN 90 code [includes b2plot]

Eirene 324 subroutines, 105007 lines of FORTRAN 90 code

Carre 102 subroutines, 18740 lines of FORTRAN 77 code

DG 41 subroutines, 34618 lines of C code

b2plot 122 subroutines, 58524 lines of FORTRAN 77 code

What resources do I need?

- a UNIX system
 - a Windows version exists but is not supported by Garching
 - * developed by the St. Petersburg group
 - * can probably contract with them for support
- a FORTRAN 90 compiler
- NAG numerical library
- NCAR graphics library
- 1 – 10 GBytes of disk space
- 128 – 512 MBytes of RAM

How do I obtain the code?

at JET a number of possibilities:

- create a “shadow tree” from `/home/dcoster/solps/solps5.0`
- use “rsync” to copy somebody’s existing tree
- other methods yet to be determined

outside of JET permission from IPP will need to be obtained, and then via

- AFS
- rsync

What resources are available?

- SOLPS web site: <http://www.ipp.mpg.de/~dpc/solps.html>
- SOLPS manual: <http://www.ipp.mpg.de/~dpc/solps/solps.pdf> or, at JET, off of <http://users.jet.efda.org/pages/edge-techgrp/index.htm>
- Eirene web site: <http://www.eirene.de/>
- two mailing lists, one for users and one for developers, described at <http://www.ipp.mpg.de/~dpc/solps.html>
- this course



```
SOLPSTOP
+---src
|   +---agg
|   |   +---IBMaix
|   |   +---data
|   |   +---sun5
|   +---aggj
|   |   +---config
|   |   +---linux_pgf90
|   |   +---src
|   +---Sonnet
|   |   +---device
|   |   |   +---upgrade.dg
+---doc
|   +---solps
|   |   +---D
|   |   +---D+C+He
```

- the `input.dat` file is unchanged
- `b2.parameters` needs to be split into
 - `b2.boundary.parameters`
 - `b2.neutrals.parameters`
 - `b2.transport.parameters`
 - `b2.numerics.parameters`
 - `b2mn.dat`
- it is probably not worth trying to restart from the B2SXDR file
- `b2plot` is largely unchanged, except for
 - `0 ... NX+1` becomes `-1 ... NX`
 - `0 ... NY+1` becomes `-1 ... NY`
 - `1 ... NS` becomes `0 ... NS-1`

Currently two versions of Eirene are available, old and new, with a third version to be installed shortly. My recommendation at the moment (and Detlev Reiter might disagree), is to use the old version unless you need features that are only in the new version.

To switch to the new version

```
sei ; gmake new ; rm LISTOBJ ; gmake clean ; gmake listobj ; gmake depend ; gmake
```

To run a case in the foreground

```
b2run b2mn > run.log
```

or

```
b2run b2mn < input.dat > run.log
```

If the 10 cpu-minute limit is a problem, you will need to use the batch system:

```
jetsubmit
```

The basic way to do a plot is

```
echo "logf 3 ndec a4p writ stdl glob rsan pvol 5 5 zrng surf  
'JET/KT1V.chords' chor 5 312 1 nspe surf 'JET/KT1V.chords' chor  
m/ surf" | b2plot
```

(all on one line)

See the examples in the analysis part.

The recommended way at JET to create a completely new tree would be to

```
mkdir ~/solps5.0_whatever  
cd ~/solps5.0_whatever  
~dcoster/solps/solps5.0/src/bin/common/lndr -c ~dcoster/solps/solps5.0 .
```

This creates a “shadow tree” where you have

- your own directory structure
- with most files as links to the MASTER (dcoster/solps/solps5.0)
- and with some directories also linked to the master

Two commands are used to replace a link by something real

gtfl replaces its arguments (which should be links to files) with copies of the files

gtldr replaces a link to a directory with a shadow tree of that directory

How can one modify the source code if the Eirene and B2 sources are not owned by you?

1. **Not recommended!** Use gtdr to get a local copy of the source directories
2. **Recommended!** Copy the relevant source file into src.local, modify, redo `gmake depend`; `gmake`

`sei ; gmake tags ; emacs`

or

`sbb ; gmake tags ; emacs`

Then, in emacs, to

goto a subroutine ESC . subroutine_name

search all the source for some text ESC x tags-search whatever